

From: ["Maheswaran, Sridevi" <Sridevi.Maheswaran@fisglobal.com>](mailto:Sridevi.Maheswaran@fisglobal.com)
To: ["Rangam, VamshiKrishna" <VamshiKrishna.Rangam@fisglobal.com>](mailto:VamshiKrishna.Rangam@fisglobal.com)
Date: 2/20/2026 8:50:55 AM
Subject: Fw: NAB - Incident Discussion

Regards,
Sridevi Maheswaran

Upcoming out of office : Feb 25 & Feb 26

From: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>
Sent: Friday, February 20, 2026 12:04 AM
To: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>; Priscilla Remo (External) <Priscilla.Remo@nab.com.au>
Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>
Subject: Re: NAB - Incident Discussion

This is great news, Mohan. I want to document few things we discovered in yesterday's (details below)

To the question on how the application works with certificates being absent: certificates are not absent. We established the fact that the NAB's root and intermediate certificates are present in the Java's cacerts file on DEV and PROD. As of now, for DataNavigator SSL communication between web and backend server - it is a one-way SSL validation - where the client has to only present the root certificate to the backend as they run in the same DMZ. Two-way SSL validation where the client presents the private certificate to the backend for validation is not implemented. That is why SSL communication works between DNWeb and backend DNServer with the NAB root certificate present in java truststore.

Back to the original root cause:

The root certificate having SHA1 is the issue here as many OS level crypto policies by default deny SHA1
So you might want to engage your security/certificate team to see if the root certificates can be reissued at a stronger cipher level like SHA256 or SHA384.

 Summary

Component	SHA-256	SHA-256	SHA-1
Leaf cert	OK		
Intermediate cert		OK	
Root cert			✖ Problem

Yes — a SHA-1 root is enough to break the TLS handshake unless you explicitly allow SHA-1.

Your observed behavior matches this perfectly.

If you want, send me the full `openssl s_client -msg` output and I can highlight the exact handshake step where SHA-1 is causing the rejection.

Likely root causes (ranked)

1. Server (or TLS-terminating device) is TLS 1.3-intolerant / misconfigured

The peer advertises TLS 1.3 in `ServerHello` and then immediately bails. This is common with older ADCs/WAFs/SSL-inspection middleboxes or older stacks that *advertise* 1.3 but don't implement the post-`ServerHello` flow correctly.

2. Certificate/signature constraints that are incompatible with TLS 1.3

TLS 1.3 **does not allow** SHA-1 for handshake signatures. If the server can only produce `rsa_pkcs1_sha1` (or is pinned to legacy algorithms), it will abort after `ServerHello`. Your environment already shows SHA-1 at the **root**; while a SHA-1 root can still be present in the *chain*, the **handshake signature** in TLS 1.3 must be SHA-256+; older stacks often fail here. (Modern security configs in Java/OpenSSL/OS crypto policies disable SHA-1 by default.)

 Check Image_Micr Ver...

3. Middlebox interference

If there's SSL inspection or a load balancer in the path, it might pick TLS 1.3 but then speak TLS 1.2 semantics (sending a clear-text alert), leading to `bad record type`.

🔧 What in Linux/OpenSSL is forcing this behavior?

System-level enforcement candidates:

Layer	What it does
<code>/etc/crypto-policies/*</code>	Disables SHA1, weak ciphers, weak DH groups
<code>/etc/ssl/openssl.cnf</code>	May include system-level restrictions
OpenSSL 3.x	Removes many legacy algorithms by default
FIPS mode	Blocks SHA1 signatures completely
GnuTLS-based apps	Also reject SHA1 unless overridden

Your symptoms exactly match a system where **SHA1 is disabled** and the server only supports SHA1-signed certs or SHA1 signature algorithms.

Thanks,
Saran

Saravanan Pichandy
IT Architect Specialist
DataNavigator
C: 414.218.6701
E: Saravanan.S.Pichandy@fisglobal.com

From: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>

Sent: Thursday, February 19, 2026 4:01 AM

To: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>; Priscilla Remo (External) <Priscilla.Remo@nab.com.au>

Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>

Subject: RE: NAB - Incident Discussion

Hi Saravanan,

Thanks for your time today and nice talking to you !!

TLS connection has been established successfully now. The root cause is the Linux policy file was suppressing the SHA1 algorithm from being sent out in the Signatures to the Server. So, the policy file was corrected to be in sync with Production, the server restarted and when the server came up, the Catalina logs shows that the handshake is successful [SHA1 sent as a part of Signatures] and rest falls in the same order of the application's function as earlier. So, sorry couldn't bother you [and CoPi] by sending the Catalina log for analysis \u-10179 ?\u-8694 ?.

Screenshot below please.

```

2026-02-19 19:57:40 [main] INFO com.fis.dnweb.servlet.DNWebInit - []-Successfully acquired lock. Beginning resource bundle generation.
2026-02-19 19:57:40 [main] INFO c.f.d.b.AbstractResourceBundleGenerator - []-Output Folder Name for Bundles: /opt/tomcat9-va/webapps/DNWebUI#VA/WEB-INF/classes/bundles/
javax.net.ssl|WARNING|01|main|2026-02-19 19:57:41.318 AEDT|ServerNameExtension.java:261|Unable to indicate server name
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.318 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: server_name
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.318 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: status_request
javax.net.ssl|INFO|01|main|2026-02-19 19:57:41.377 AEDT|AlpnExtension.java:178|No available application protocols
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.378 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: application_layer_protocol_negotiation
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.378 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: status_request_v2
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.378 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: renegotiation_info
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.381 AEDT|ClientHello.java:564|Produced ClientHello handshake message (
"ClientHello": {
  "client version" : "TLSv1.2",
  "random" : "A1 EF 86 77 F9 75 22 03 13 6F B6 A1 FE 27 08 60 9C D4 6A EF 9E 35 C5 0E F5 F1 97 57 A4 EF B7 6A",
  "session id" : "",
  "cipher suites" : "[TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384(0xc02c), TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256(0xc02b), TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384(0xc030), TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256(0xc02f), TLS_DHE_RSA_WITH_AES_256_GCM_SHA384(0x0009F), TLS_DHE_RSA_WITH_AES_128_GCM_SHA256(0x0009E), TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384(0xc024), TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384(0xc028), TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256(0xc023), TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256(0xc027), TLS_DHE_RSA_WITH_AES_256_CBC_SHA256(0x0068), TLS_DHE_RSA_WITH_AES_128_CBC_SHA256(0x0067), TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA(0xc00a), TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA(0xc014), TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA(0xc009), TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA(0xc013), TLS_DHE_RSA_WITH_AES_256_CBC_SHA(0x0039), TLS_DHE_RSA_WITH_AES_128_CBC_SHA(0x0033), TLS_RSA_WITH_AES_256_GCM_SHA384(0x009d), TLS_RSA_WITH_AES_128_GCM_SHA256(0x009c), TLS_RSA_WITH_AES_256_CBC_SHA256(0x003d), TLS_RSA_WITH_AES_128_CBC_SHA256(0x003c), TLS_RSA_WITH_AES_256_CBC_SHA(0x0035), TLS_RSA_WITH_AES_128_CBC_SHA(0x002f), TLS_EMPTY_RENEGOTIATION_INFO_SCSV(0x00ff)]",
  "compression methods" : "00",
  "extensions" : {
    "supported_groups (10)": {
      "versions": [secp256r1, secp384r1, secp521r1, ffdhe2048, ffdhe3072, ffdhe4096, ffdhe6144, ffdhe8192]
    },
    "ec_point_formats (11)": {
      "formats": [uncompressed]
    },
    "signature_algorithms (13)": {
      "signature schemes": [ecdsa_secp256r1_sha256, ecdsa_secp384r1_sha384, ecdsa_secp521r1_sha512, rsa_pss_rsae_sha256, rsa_pss_rsae_sha384, rsa_pss_rsae_sha512, rsa_pss_pss_sha256, rsa_pss_pss_sha384, rsa_pss_pss_sha512, rsa_pkcs1_sha256, rsa_pkcs1_sha384, rsa_pkcs1_sha512, dsa_sha256, ecdsa_sha224, rsa_sha224, dsa_sha224, ecdsa_sha1, rsa_pkcs1_sha1, dsa_sha1]
    },
    "signature_algorithms_cert (50)": {
      "signature schemes": [ecdsa_secp256r1_sha256, ecdsa_secp384r1_sha384, ecdsa_secp521r1_sha512, rsa_pss_rsae_sha256, rsa_pss_rsae_sha384, rsa_pss_rsae_sha512, rsa_pss_pss_sha256, rsa_pss_pss_sha384, rsa_pss_pss_sha512, rsa_pkcs1_sha256, rsa_pkcs1_sha384, rsa_pkcs1_sha512, dsa_sha256, ecdsa_sha224, rsa_sha224, dsa_sha224, ecdsa_sha1, rsa_pkcs1_sha1, dsa_sha1]
    },
    "extended_master_secret (23)": {
      <empty>
    },
    "supported_versions (43)": {
      "versions": [TLSv1.2]
    }
  }
}
)
javax.net.ssl|FINE|01|main|2026-02-19 19:57:41.424 AEDT|ServerHello.java:863|Consuming ServerHello handshake message (
"ServerHello": {
  "server version" : "TLSv1.2",
  "random" : "69 96 D0 85 F2 3E 4E 17 7A 56 6A 15 C9 FB 1C D9 0C BC AB A2 66 7C E1 79 44 4F 57 4E 47 52 44 01",
  "session id" : "00 00 03 1F 0A D1 8E 44 60 EC 00 00 00 00 00 00 00 00 00 00 00 69 96 D0 85 00 00 08 2A",
  "cipher suite" : "TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384(0xc030)",
  "compression methods" : "00",
  "extensions" : {
    "extended_master_secret (23)": {
      <empty>
    },
    "ec_point_formats (11)": {
      "formats": [uncompressed]
    },
    "renegotiation_info (65,281)": {
      "renegotiated connection": [<no renegotiated connection>]
    }
  }
}

```

And the bundles downloaded exactly at that time as expected and the GUI navigates as usual !


```

-rw-r----- 1 tomcat tomcat 831 Feb 19 19:58 NABT_ACCOUNT_TYPE.properties
-rw-r----- 1 tomcat tomcat 7686 Feb 19 19:58 NABT_ACTION_CODE.properties
-rw-r----- 1 tomcat tomcat 60259 Feb 19 19:58 NABT_BUSINESS_CODE.properties
-rw-r----- 1 tomcat tomcat 7259 Feb 19 19:58 NABT_CNTRY_CODE_2.properties
-rw-r----- 1 tomcat tomcat 7512 Feb 19 19:58 NABT_CNTRY_CODE_A.properties
-rw-r----- 1 tomcat tomcat 7034 Feb 19 19:58 NABT_CNTRY_CODE.properties
-rw-r----- 1 tomcat tomcat 18810 Feb 19 19:58 NABT_CURRENCY_CODE.properties
-rw-r----- 1 tomcat tomcat 39831 Feb 19 19:58 NABT_DEVICE_STATUS.properties
-rw-r----- 1 tomcat tomcat 173 Feb 19 19:58 NABT_DV_REASON_CODE.properties
-rw-r----- 1 tomcat tomcat 25695 Feb 19 19:58 NABT_EMSR_ACTION.properties
-rw-r----- 1 tomcat tomcat 4140 Feb 19 19:58 NABT_EMSR_PHASE.properties
-rw-r----- 1 tomcat tomcat 51671 Feb 19 19:58 NABT_EMSR_STATE.properties
-rw-r----- 1 tomcat tomcat 3058 Feb 19 19:58 NABT_EMSR_STATUS.properties
-rw-r----- 1 tomcat tomcat 34646 Feb 19 19:58 NABT_ERROR_MSGS.properties
-rw-r----- 1 tomcat tomcat 376 Feb 19 19:58 NABT_FEE_TYPE.properties
-rw-r----- 1 tomcat tomcat 2156 Feb 19 19:58 NABT_FIN_TYPE.properties
-rw-r----- 1 tomcat tomcat 2557 Feb 19 19:58 NABT_FUNCTION_CODE.properties
-rw-r----- 1 tomcat tomcat 1678 Feb 19 19:58 NABT_FUNCTION_TYPE.properties
-rw-r----- 1 tomcat tomcat 18683 Feb 19 19:58 NABT_MC_DATAELEMENT.properties
-rw-r----- 1 tomcat tomcat 95798 Feb 19 19:58 NABT_MC_ERROR.properties
-rw-r----- 1 tomcat tomcat 269028 Feb 19 19:58 NABT_MISC_CONFIG.properties
-rw-r----- 1 tomcat tomcat 11888 Feb 19 19:58 NABT_MSG_REASON.properties
-rw-r----- 1 tomcat tomcat 9454 Feb 19 19:58 NABT_NETWORK_ID.properties
-rw-r----- 1 tomcat tomcat 65107 Feb 19 19:58 NABT_TRAN_TYPE_IND.properties
-rw-r----- 1 tomcat tomcat 189995 Feb 19 19:58 NABT_VROL_ERR_MSG.properties
[root@dnr1idw1001 logs]#

```

But still good to have clarity [as will help us to maintain the application bit efficiently] on how the application works despite the certificates being absent in Tomcat stores but maybe for another time.

Sorry for holding you late. Have a nice day !!

Hi [@Uma.Maheswari](#)/[@Jadadeli, Priya](#) – Kindly close & remove this “Zero Length Response from DN server” issue from our weekly list as this is a connectivity problem that is resolved now. Thanks.

Kind Regards,
Mohan

From: Mohan Prasath Manickam

Sent: Wednesday, 18 February 2026 3:20 PM

To: 'Pichandy, Saravanan S' <Saravanan.S.Pichandy@fisglobal.com>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; Oleksandr.Pysarenko <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <Marco.Allori@fisglobal.com>; Priscilla Remo (External) <Priscilla.Remo@nab.com.au>

Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>

Subject: RE: NAB - Incident Discussion

Hi Saravanan,

I couldn't see the DN certificate there. Thanks.

Below from Prod cert:

```
Alias name: prod-dnr1pw1001-ssl-03
Creation date: Jul 10, 2025
Entry type: PrivateKeyEntry
Certificate chain length: 3
Certificate[1]:
Owner: CN=dnr1pw1001.bas.aur.national.com.au, OU=Products-Acquiring Technology Prod Supp, O=National Australia Bank Limited, L=Melbourne, ST=Victoria, C=AU
Issuer: CN=NAB Internal Issuing AD CA - KNOX P01, DC=aur, DC=national, DC=com, DC=au
```

And when I used your command, and grep'ed for "Owner: CN=dnr", there are no occurrence of the string [Owner: CN=dnr1pw1001.bas.aur.national.com.au].

```
Owner: CN=COMODO RSA Certi
Owner: CN=D-TRUST BR Root
Owner: CN=D-TRUST BR Root
Owner: CN=D-TRUST EV Root
Owner: CN=D-TRUST EV Root
Owner: CN=D-TRUST Root Cla
Owner: CN=D-TRUST Root Cla
Owner: CN=DigiCert Assured
Owner: CN=DigiCert Assured
Owner: CN=DigiCert Assured
Owner: CN=DigiCert Global
Owner: CN=DigiCert Global
Owner: CN=DigiCert Global
Owner: CN=DigiCert High As
Owner: CN=DigiCert TLS ECC
Owner: CN=DigiCert TLS RSA
Owner: CN=DigiCert Trusted
Owner: CN=e-Signo Root CA
```

Kind Regards,
Mohan

From: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>

Sent: Wednesday, 18 February 2026 2:47 PM

To: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; Oleksandr.Pysarenko <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <Marco.Allori@fisglobal.com>; Priscilla Remo (External) <Priscilla.Remo@nab.com.au>

Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>

Subject: Re: NAB - Incident Discussion

CAUTION: This email originated from outside of the organisation. Do not act on instructions, click links, open attachments or scan QR codes unless you recognise the sender and know the content is authentic and safe.

Hi Mohan -

Can you try listing the certificates in "cacerts" and see if the DNServer certificate is listed

Command:
keytool -list -v -keystore <path_to_cacerts> -storepass changeit

Thanks,
Saran

Saravanan Pichandy
IT Architect Specialist
DataNavigator
C: 414.218.6701
E: Saravanan.S.Pichandy@fisglobal.com
FIS | Empowering the Financial World    

From: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>
Sent: Tuesday, February 17, 2026 9:36 PM
To: Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>; Priscilla Remo (External) <Priscilla.Remo@nab.com.au>
Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>
Subject: RE: NAB - Incident Discussion

Thanks Uma !! Should be fine as the meeting time is immediately after our weekly Jira catchup.

Hi Saravanan,

I guess, I am referring to the right location on Prod for the Trust store. And I don't see any DN custom certificates in there.

```
[root@dnrlipw1001 logs]# JRE=$(dirname $(dirname $(readlink -f /usr/bin/java)))
[root@dnrlipw1001 logs]# echo "$JRE/lib/security/cacerts"
/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/jre/lib/security/cacerts
[root@dnrlipw1001 logs]# ll /usr/lib/jvm/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/jre/lib/security/
total 172
lrwxrwxrwx 1 root root      105 Jan 21 14:35 blacklisted.certs -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/blacklisted.certs
lrwxrwxrwx 1 root root       95 Jan 21 14:35 cacerts -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/cacerts
-r--r--r-- 1 root root 174348 Jan 19 09:34 cacerts.upstream
lrwxrwxrwx 1 root root       99 Jan 21 14:35 java.policy -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/java.policy
lrwxrwxrwx 1 root root      101 Jan 21 14:35 java.security -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/java.security
lrwxrwxrwx 1 root root       95 Jan 21 14:35 nss.cfg -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/nss.cfg
lrwxrwxrwx 1 root root      100 Jan 21 14:35 nss.fips.cfg -> /etc/java/java-1.8.0-openjdk/java-1.8.0-openjdk-1.8.0.482.b08-1.el8.x86_64/lib/security/nss.fips.cfg
drwxr-xr-x 4 root root       38 Jan 19 09:34 policy
```

Kind Regards,
Mohan

From: Maheswari, Uma <Uma.Maheswari@fisglobal.com>

Sent: Wednesday, 18 February 2026 1:58 PM

To: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>; Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Oleksandr.Pysarenko <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <Marco.Allori@fisglobal.com>

Cc: Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>; au dn support <au.dn.support@fisglobal.com>

Subject: RE: NAB - Incident Discussion

CAUTION: This email originated from outside of the organisation. Do not act on instructions, click links, open attachments or scan QR codes unless you recognise the sender and know the content is authentic and safe.

Thanks [@Pichandy, Saravanan S](#).

[@Mohan Prasath Manickam](#) - I have shared the invite. Please forward to anyone else if required.

Let us know if this time does not work.

Thanks, Uma

From: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>

Sent: 18 February 2026 08:18

To: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>

Cc: Allori, Marco <Marco.Allori@fisglobal.com>; Ponnambalam, Ganesh <Ganesh.Ponnambalam@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>

Subject: Re: NAB - Incident Discussion

Hi Mohan -

Can you also inspect the Production Java truststore and see if the DNServer certificate has been imported there?

[@Maheswari, Uma](#) - please setup a call for Wednesday 2/18 10.30PM US CST (Thursday 10AM IST / Thursday 3.30PM AEDT). If this time does not work, please let me know.

Thanks,
Saran

Saravanan Pichandy
IT Architect Specialist



From: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>

Sent: Tuesday, February 17, 2026 8:15 PM

To: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>

Subject: RE: NAB - Incident Discussion

Hi,

I was looking at how Prod is configured before applying on Dev. I see that the same certificate used by the browser to Tomcat is being used for Tomcat to Mainframes. A snip from server.xml below:

root@dnrlipw1001:/opt/tomcat/logs

```
<!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443
This connector uses the NIO implementation. The default
SSLImplementation will depend on the presence of the APR/native
library and the useOpenSSL attribute of the AprLifecycleListener.
Either JSSE or OpenSSL style configuration may be used regardless of
the SSLImplementation selected. JSSE style configuration is used below.
-->
<!--
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
    maxThreads="150" SSLEnabled="true"
    maxParameterCount="1000"
    >
    <SSLHostConfig>
        <Certificate certificateKeystoreFile="conf/localhost-rsa.jks"
            certificateKeystorePassword="changeit" type="RSA" />
    </SSLHostConfig>
</Connector>
-->
<Connector port="8443" protocol="HTTP/1.1"
    maxThreads="300" SSLEnabled="true" scheme="https" secure="true"
    clientAuth="false" sslProtocol="TLS"
    sslEnabledProtocols="TLSv1,TLSv1.1,TLSv1.2"
    keystoreFile="/opt/tomcat/cert/dnrcert.jks"
    keystorePass="1bu1h@gajeTR1!" server=" " />
<Connector port="10000" protocol="HTTP/1.1"
    maxThreads="300" SSLEnabled="true" scheme="https" secure="true"
    clientAuth="false" sslProtocol="TLS"
    sslEnabledProtocols="TLSv1,TLSv1.1,TLSv1.2"
    keystoreFile="/opt/tomcat/cert/dnrcert.jks"
    keystorePass="1bu1h@gajeTR1!" server=" " />
<!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443 with HTTP/2
This connector uses the APR/native implementation which always uses
OpenSSL for TLS.
Either JSSE or OpenSSL style configuration may be used. OpenSSL style
configuration is used below.
-->
<!--
<Connector port="8443" protocol="org.apache.coyote.http11.Http11AprProtocol"
    maxThreads="150" SSLEnabled="true"
    maxParameterCount="1000"
    >
    <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />
    <SSLHostConfig>
        <Certificate certificateKeyFile="conf/localhost-rsa-key.pem"
            certificateFile="conf/localhost-rsa-cert.pem"
            certificateChainFile="conf/localhost-rsa-chain.pem"
            type="RSA" />
    </SSLHostConfig>
</Connector>
-->
```

Apart from that, I have few other queries that needs clarification. So would like to organise a quick 30 mins. Can you please let me know whom all should be in the invite ? Thanks.

Kind Regards,

Mohan

From: Mohan Prasath Manickam

Sent: Wednesday, 18 February 2026 10:33 AM

To: 'Pichandy, Saravanan S' <Saravanan.S.Pichandy@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Oleksandr.Pysarenko <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <Marco.Allori@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>

Subject: RE: NAB - Incident Discussion

Hi Saravanan,

We haven't done that yet. Will try and get back. Thanks.

Kind Regards,
Mohan

From: Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>

Sent: Wednesday, 18 February 2026 3:26 AM

To: Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Oleksandr.Pysarenko <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <Marco.Allori@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>

Subject: Re: NAB - Incident Discussion

You don't often get email from saravanan.s.pichandy@fisglobal.com. [Learn why this is important](#)

CAUTION: This email originated from outside of the organisation. Do not act on instructions, click links, open attachments or scan QR codes unless you recognise the sender and know the content is authentic and safe.

Hi -
Has the new certificate from DNServer imported into Java truststore that Tomcat is running?

If not, please import the new certificate from DNServer into the Java truststore - example import command is:

```
/opt/java/bin/keytool -importcert -alias dev_dnserver_cert -keystore "/opt/java/jre/lib/security/cacerts" -file /opt/FIS/workspace/DNCertificate.pem
```

And then bounce Tomcat.

If this has already been done, please setup a working session to further research the issue. [@Maheswaran, Sridevi](#) - we might need server resources on the working session to generate/review CI traces.

Thanks,
Saran

Saravanan Pichandy

IT Architect Specialist

DataNavigator

C: 414.218.6701

E: Saravanan.S.Pichandy@fisglobal.com

FIS | Empowering the Financial World    

From: Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>

Sent: Tuesday, February 17, 2026 10:17 AM

To: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>; Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>; Pichandy, Saravanan S <Saravanan.S.Pichandy@fisglobal.com>; Cincotta, Tim <Tim.Cincotta@fisglobal.com>; Narayanan, Sivaraj <Sivaraj.Narayanan@fisglobal.com>
Subject: RE: NAB - Incident Discussion

Hi Saran, Tim –

Please review the below query and provide your inputs to NAB.

Regards,
Shakir

From: Mohan Prasath Manickam <Mohan.P.Manickam@nab.com.au>

Sent: Tuesday, February 17, 2026 3:34 PM

To: Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Maheswari, Uma <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Allori, Marco <Marco.Allori@fisglobal.com>

Subject: RE: NAB - Incident Discussion

Hi FIS Team,

Further to my earlier email [on "Zero Length Response from DN server"], we are trying to enable TLS in one of our Dev regions [VA]. We did the below stuff in conjunction to Production:

1. Got a new certificate [exclusive keyring to VA] installed on Mainframes.
2. Secured the VA region's Mainframe socket.
3. Enabled the TLS parms [TLSecuredPorts & LimitTlsVersions] in the JSON file.

Then initiated the connection, but when the application tries to hit the server, it fails at the handshake level which results in a RC-467 at the TLS side which translates to "Signature algorithm not in signature algorithm pairs list".

There are 2 mandatory signatures the Server expects (i) SHA1 & (ii) SHA256. When we traced the handshake to see which one is missing, it points to SHA1. Screen shot as below:


```

2026-02-17 17:47:58 [https-jsse-nio-10002-exec-2] INFO com.fis.dn.services.LogonService - [SYSTEM]-calling the logon service...
javax.net.ssl|WARNING|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.259 AEDT|ServerNameExtension.java:261|Unable to indicate server name
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.259 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: server_name
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.259 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: status_request
javax.net.ssl|INFO|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.260 AEDT|AlpnExtension.java:178|No available application protocols
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.260 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: application_layer_protocol_negotiation
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.260 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: status_request_v2
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.260 AEDT|SSLExtensions.java:260|Ignore, context unavailable extension: renegotiation_info
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.261 AEDT|ClientHello.java:564|Produced ClientHello handshake message (
"ClientHello": {
  "client_version" : "TLSv1.2",
  "random" : "0C 2D 58 24 9C 77 88 6A 87 70 5C E0 90 61 FA 46 A6 6E 81 66 88 7A AC 77 13 36 07 8E E1 00 CD 50",
  "session_id" : "",
  "cipher_suites" : "[TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384(0xc02c), TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256(0xc02b), TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384(0xc030), TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256(0xc02f), TLS_DHE_RSA_WITH_AES_256_GCM_SHA384(0xc00f), TLS_DHE_RSA_WITH_AES_128_GCM_SHA256(0xc00e), TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384(0xc024), TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256(0xc023), TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256(0xc027), TLS_DHE_RSA_WITH_AES_256_CBC_SHA256(0xc006), TLS_DHE_RSA_WITH_AES_128_CBC_SHA256(0xc007), TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA(0xc00a), TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA(0xc014), TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA(0xc009), TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA(0xc011), TLS_DHE_RSA_WITH_AES_256_CBC_SHA(0xc039), TLS_DHE_RSA_WITH_AES_128_CBC_SHA(0xc033), TLS_RSA_WITH_AES_256_GCM_SHA384(0xc030), TLS_RSA_WITH_AES_128_GCM_SHA256(0xc03c), TLS_RSA_WITH_AES_256_CBC_SHA256(0xc03d), TLS_RSA_WITH_AES_128_CBC_SHA256(0xc03e), TLS_RSA_WITH_AES_256_GCM_SHA384(0xc03f), TLS_EMPTY_RENEGOTIATION_INFO_SCSV(0xc00f)]",
  "compression_methods" : "00",
  "extensions" : {
    "supported_groups (10)": {
      "versions": [secp256r1, secp384r1, secp521r1, ffdhe2048, ffdhe3072, ffdhe4096, ffdhe6144, ffdhe8192]
    },
    "ec_point_formats (11)": {
      "formats": [uncompressed]
    },
    "signature_algorithms (13)": {
      "signature_schemes": [rsa_pkcs1_sha256, rsa_pkcs1_sha384, rsa_pkcs1_sha512]
    },
    "signature_algorithms_cert (50)": {
      "signature_schemes": [rsa_pkcs1_sha256, rsa_pkcs1_sha384, rsa_pkcs1_sha512]
    },
    "extended_master_secret (23)": {
      "empty": true
    },
    "supported_versions (43)": {
      "versions": [TLSv1.2]
    }
  }
}
)
javax.net.ssl|FINE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.283 AEDT|Alert.java:238|Received alert message (
Alert: {
  "level": "fatal",
  "description": "handshake_failure"
}
)
javax.net.ssl|SEVERE|22|https-jsse-nio-10002-exec-2|2026-02-17 17:47:58.284 AEDT|TransportContext.java:323|Fatal (HANDSHAKE_FAILURE): Received fatal alert: handshake_failure (
throwable: {
  javax.net.ssl.SSLHandshakeException: Received fatal alert: handshake_failure
}
)

```

Comparing Dev to Prod [working TLS connection], I guess [haven't traced so can't confirm 100%], both the signatures are being sent [crosschecked by forcing TLS connection without SHA1 or SHA256 and handshake fails].

So, bit confused on why tomcat doesn't send in Dev but in Prod. Can you please advise what could be the issue? Thanks.

Note – "Zero Length Response from DN server" persists as there is no response from Server because of incomplete connection.

Kind Regards,
Mohan

From: Mohan Prasath Manickam

Sent: Monday, 2 February 2026 4:54 PM

To: 'Pysarenko, Oleksandr' <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; 'Maheswaran, Sridevi' <Sridevi.Maheswaran@fisglobal.com>; 'Shaik, Shakir Ahamamd' <Shaik.S.Ahamamd@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>; Peter L Williams <Peter.L.Williams@nab.com.au>; Marco.Allori <marco.allori@fisglobal.com>
Subject: RE: NAB - Incident Discussion

Hi All,

We were investigating in parallel on the "Zero Length Response from DN server" issue and seems to be a bit of a gap in DN non-prod and prod environment's TLS configuration.

Basically, in the non-prod envs, the TLS is not enabled on the comms port between the Tomcat server and Mainframes DN Server. So, whenever the "TLS" parameters are added in the Json file, Tomcat tries to initiate a secured connection with DN Mainframe server which it doesn't understand and so, it closes the connection immediately [after this, the actual request would be passed to the DN server and a response sent back to Tomcat]. Because of this no response scenario, Tomcat gets a zero-length error. We are in the process of getting the configs in order now.

```
2025-11-24 16:23:43 [main] INFO com.fis.dnweb.servlet.DNWebInit - {}-Successfully acquired lock. Beginning resource bundle generation.
2025-11-24 16:23:43 [main] INFO c.f.d.b.AbstractResourceBundleGenerator - {}-Output Folder Name for Bundles: /opt/tomcat9-sit5/webapps/DNWebUI#SIT5/WEB-INF/classes/D
2025-11-24 16:23:44 [main] INFO c.f.d.b.AbstractResourceBundleGenerator - {}-Attempting login to switch: ****
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-Checking for External configuration properties file ...
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-... found :/opt/security/apps/dn/overrides/dnwebsit5-json-overrides.json
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: dn/application.properties
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: dn/userpreferences.defaults
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: application.properties
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: databaseConnection.properties
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: encryption.properties
2025-11-24 16:23:44 [main] INFO c.f.c.r.properties.AppProperties - {}-File Name: report-processor.properties
2025-11-24 16:23:44 [main] INFO com.fis.dn.services.LoginService - {}-Calling the login service...
2025-11-24 16:23:44 [main] ERROR c.f.c.s.SoapServiceComponent - {}-Zero length response from server
java.lang.IllegalStateException: Zero length response from server
at com.fis.commons.baseconnector.BaseCommunicator.getResponse(BaseCommunicator.java:257)
```

So, kindly hold this for a bit until we correct the configuration stuff, test & confirm back. Thanks.

Kind Regards,
Mohan

From: Mohan Prasath Manickam
Sent: Tuesday, 27 January 2026 7:43 PM
To: 'Pysarenko, Oleksandr' <Oleksandr.Pysarenko@fisglobal.com>; Jadadeli, Priya <Priya.Jadadeli@fisglobal.com>; Uma.Maheswari <Uma.Maheswari@fisglobal.com>; Maheswaran, Sridevi <Sridevi.Maheswaran@fisglobal.com>; Shaik, Shakir Ahamamd <Shaik.S.Ahamamd@fisglobal.com>; Stephen W Antonopoulos <Stephen.W.Antonopou@nab.com.au>
Subject: RE: NAB - Incident Discussion

Hi Sridevi,

As asked, please find the Catalina logs [for today] where we tried adding the TLS parameters ["key": "TLSSecuredPorts" & "key": "LimitTlsVersions"] to the override JSON file. Thanks.

Kind Regards,
Mohan

-----Original Appointment-----

From: Pysarenko, Oleksandr <Oleksandr.Pysarenko@fisglobal.com>

Sent: Thursday, 22 January 2026 11:23 AM

To: Pysarenko, Oleksandr; Mohan Prasath Manickam; Priya.Jadadeli; Uma.Maheswari; Maheswaran, Sridevi; Shaik, Shakir Ahamamd

Subject: NAB - Incident Discussion

When: Tuesday, 27 January 2026 4:30 PM-5:00 PM (UTC+10:00) Canberra, Melbourne, Sydney.

Where: Microsoft Teams Meeting

CAUTION: This email originated from outside of the organisation. Do not act on instructions, click links, open attachments or scan QR codes unless you recognise the sender and know the content is authentic and safe.

Hi All,

Scheduling meeting to discuss Incident Ticket with NAB shown below:

IN24013987214 : Zero Length Response from DN server

Thanks Oleksandr

Microsoft Teams meeting

Join: <https://teams.microsoft.com/meet/32654519259230?p=jsS5u8fEQb95CDb3Fm>

Meeting ID: 326 545 192 592 30

Passcode: f2ZB3yW9

[Need help?](#) | [System reference](#)

Join on a video conferencing device

Tenant key: teams@fisglobal.onpexip.com

Video ID: 125 904 912 3

[More info](#)

For organizers: [Meeting options](#)

This communication may be recorded. You will be notified if a recording starts; and may disconnect if you do not wish to be recorded.

The information contained in this message is proprietary and/or confidential. If you are not the intended recipient, please: (i) delete the message and all copies; (ii) do not disclose, distribute or use the message in any manner; and (iii) notify the sender immediately. In addition, please be aware that any message addressed to our domain is subject to archiving and review by persons other than the intended recipient. Thank you.

Report suspicious and spam emails via the Phish Alert Report button

Report suspicious NAB-branded SMS to 047 NAB 0003 (0476 220 003)

Report suspicious calls to phish@nab.com.au

Learn more at [go/groupsecurity](https://go.groupsecurity)

The information contained in this email communication may be confidential. If you have received this email in error, please notify the sender by return email, delete this email and destroy any copy.

Any advice contained in this email has been prepared without taking into account your objectives, financial situation or needs. Before acting on any advice in this email, National Australia Bank Limited (NAB) recommends that you consider whether it is appropriate for your circumstances. If this email contains reference to any financial products, NAB recommends you consider the Product Disclosure Statement (PDS) or other disclosure document available from NAB, before making any decisions regarding any products.

The information contained in this message is proprietary and/or confidential. If you are not the intended recipient, please: (i) delete the message and all copies; (ii) do not disclose, distribute or use the message in any manner; and (iii) notify the sender immediately. In addition, please be aware that any message addressed to our domain is subject to archiving and review by persons other than the intended recipient. Thank you.

Report suspicious and spam emails via the Phish Alert Report button

Report suspicious NAB-branded SMS to 047 NAB 0003 (0476 220 003)

Report suspicious calls to phish@nab.com.au

Learn more at [go/groupsecurity](https://go.groupsecurity)

The information contained in this email communication may be confidential. If you have received this email in error, please notify the sender by return email, delete this email and destroy any copy.

Any advice contained in this email has been prepared without taking into account your objectives, financial situation or needs. Before acting on any advice in this email, National Australia Bank Limited (NAB) recommends that you consider whether it is appropriate for your circumstances. If this email contains reference to any financial products, NAB recommends you consider the Product Disclosure Statement (PDS) or other disclosure document available from NAB, before making any decisions regarding any products.

The information contained in this message is proprietary and/or confidential. If you are not the intended recipient, please: (i) delete the message and all copies; (ii) do not disclose, distribute or use the message in any manner; and (iii) notify the sender immediately. In addition, please be aware that any message addressed to our domain is subject to archiving and review by persons other than the intended recipient. Thank you.

Report suspicious and spam emails via the Phish Alert Report button
Report suspicious NAB-branded SMS to 047 NAB 0003 (0476 220 003)
Report suspicious calls to phish@nab.com.au
Learn more at [go/groupsecurity](https://go.groupsecurity)

The information contained in this email communication may be confidential. If you have received this email in error, please notify the sender by return email, delete this email and destroy any copy.

Any advice contained in this email has been prepared without taking into account your objectives, financial situation or needs. Before acting on any advice in this email, National Australia Bank Limited (NAB) recommends that you consider whether it is appropriate for your circumstances. If this email contains reference to any financial products, NAB recommends you consider the Product Disclosure Statement (PDS) or other disclosure document available from NAB, before making any decisions regarding any products.

The information contained in this message is proprietary and/or confidential. If you are not the intended recipient, please: (i) delete the message and all copies; (ii) do not disclose, distribute or use the message in any manner; and (iii) notify the sender immediately. In addition, please be aware that any message addressed to our domain is subject to archiving and review by persons other than the intended recipient. Thank you.

Report suspicious and spam emails via the Phish Alert Report button
Report suspicious NAB-branded SMS to 047 NAB 0003 (0476 220 003)
Report suspicious calls to phish@nab.com.au
Learn more at [go/groupsecurity](https://go.groupsecurity)

The information contained in this email communication may be confidential. If you have received this email in error, please notify the sender by return email, delete this email and destroy any copy.

Any advice contained in this email has been prepared without taking into account your objectives, financial situation or needs. Before acting on any advice in this email, National Australia Bank Limited (NAB) recommends that you consider whether it is appropriate for your circumstances. If this email contains reference to any financial products, NAB recommends you consider the Product Disclosure Statement

(PDS) or other disclosure document available from NAB, before making any decisions regarding any products.